Earth and Planetary Science Letters, 76 (1985/86) 151–166
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[3]

Jurassic magnetostratigraphy, 2. Middle-Late Oxfordian of Aguilon, Iberian Cordillera, northern Spain

M.B. Steiner [1], J.G. Ogg [1,2], G. Melendez [3] and L. Sequeiros [4]

[1] *Department of Geology and Geophysics, University of Wyoming, Laramie, WY 82071 (U.S.A.)*
[2] *A-012, Scripps Institution of Oceanography, UCSD, La Jolla, CA 92093 (U.S.A.)*
[3] *Dep. de Paleontologia, Fac. Ciencias, Ciudad Universitaria, Zaragoza 50009 (Spain)*
[4] *Seccion de Geologia La Rabida, Palos de la Frontera, Huelva (Spain)*

Received January 21, 1985; revised version received August 30, 1985

A reproducible magnetic polarity pattern has been obtained from four overlapping stratigraphic sequences of ammonite-rich gray pelagic limestones, spanning part of the Middle and Late Oxfordian (*G. transversarium* to *E. bimammatum* ammonite zones). Progressive thermal demagnetization removed a present field overprint. Fold tests, both between- and within-outcrops, are positive. The mean direction of magnetization is 322°, 45° ($\alpha_{95} = 6°$), which implies a paleolatitude of 26°N. Frequent changes of polarity were observed. The magnetostratigraphic sequence contains the ammonite zones, subzones, and horizons of the Tethyan faunal realm. Using the reference horizons provided by this detailed ammonite zonation, the polarity zones can be readily correlated among all sections. The composite polarity pattern is predominantly of reversed polarity, punctuated by a number of relatively shorter duration normal polarity intervals. In order to compare this sequence to that of the marine magnetic anomalies, we have revised the age estimates for the pre-M25 marine magnetic normaly pattern according to the most recent time scale. Based on those ages, the sedimentary record at Aguilon suggests a much greater frequency of polarity changes than does either model derived from the two sets of marine magnetic anomalies of this time. This sedimentary record probably is a truer approximation of the actual geomagnetic polarity history because of the continuity of sedimentation in these sections and the inherent ambiguities involved in determining a polarity pattern from the very low amplitudes that everywhere characterize this part of the marine record.

1. Introduction

The history of reversals of the geomagnetic field during the Late Jurassic is currently known primarily from the marine magnetic anomaly record of the oceans, the "M sequence" [1–3]. For the Late Cretaceous and Cenozoic, the technique of technique of block modeling of marine magnetic anomalies has yielded a record of geomagnetic polarity history [4] that has proven to be fairly accurate, as has been demonstrated by magnetostratigraphic studies of sedimentary and volcanic sequences [5]. The Early Cretaceous and Late Jurassic M sequence, in part, also has been verified and dated by magnetostratigraphic studies. M0 throuh M3 have been determined to be of Aptian through Barremian age [6,7]; and M14 through M25 have been identified in sedimentary rocks of Berriasian through Kimmeridgian age [8–14]. Thus, to date, block models of the marine record have provided an accurate record of the actual geomagnetic reversal history.

Beginning with anomaly M23, the marine record displays decreasing anomaly amplitude with increasing anomaly age. The amplitude continues decreasing to the point that much of the oldest portion of the ocean floors has little or no anomaly signature. This situation gave rise to the term "Jurassic Quiet Zone" [1,2] and the concept of a long interval of normal polarity in the Middle and Late Jurassic [2,15]. Later geophysical surveys identified subdued linear anomalies within this "quiet zone". However, the pattern from the Pacific [3,16] does not match the pattern observed in the western Atlantic [17,18], as shown in Fig. 1. It appears that the relatively slower spreading rate of the Atlantic was inadequate to record the details of the polarity sequence which are resolvable in the Pacific. Recently, D. Handschumacher has revised the polarity sequence between M25 to M29

0012-821X/85/$03.30 © 1985 Elsevier Science Publishers B.V.

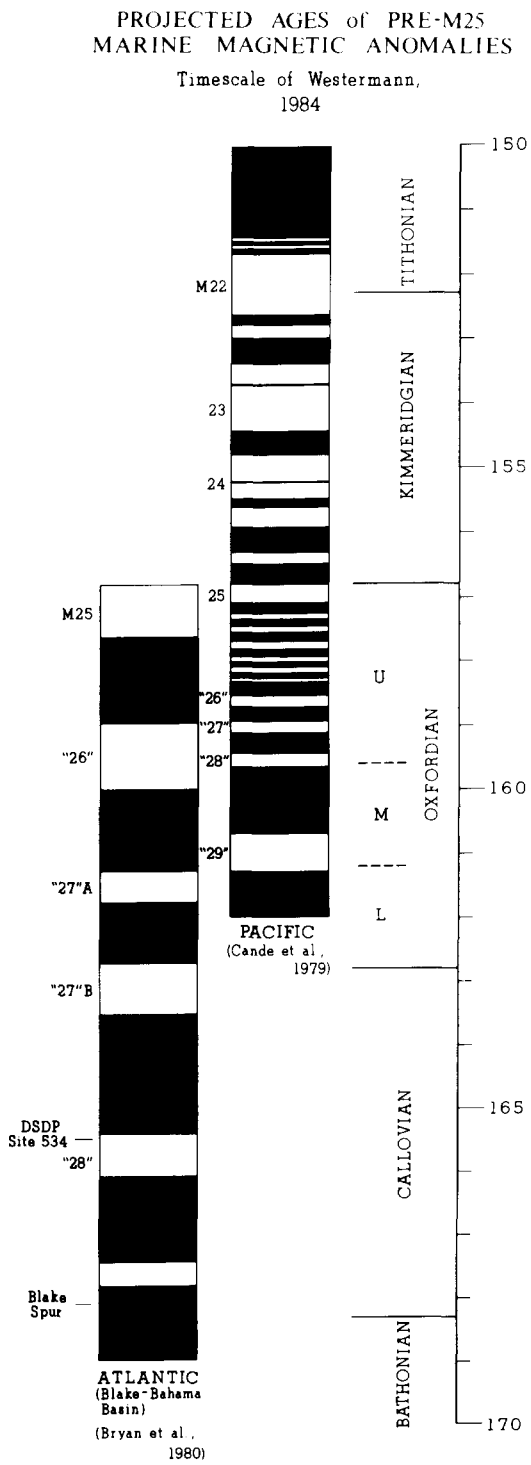


Fig. 1. Estimated ages of marine magnetic anomalies M22 through "M29" of the Pacific block model of Cande et al. [3] and M25 through "M28" of the Atlantic block model derived from Bryan et al. [18]. The biostratigraphic and absolute age estimates shown are from Westermann's [47,48] time scale for the Jurassic (see text).

[19] and has identified a pattern of anomalies in the western Pacific beyond M25 that shows very frequent polarity changes, and nearly equal amounts of normally and reversely magnetized crust (D. Handschumacher, personal communication, 1984).

A number of previous paleomagnetic studies have investigated Oxfordian age sedimentary rocks. However, none of these has produced an Oxfordian polarity sequence, either because of poor biostratigraphic control or poor magnetic results. The lower part of the Morrison Formation (western U.S.A.), of supposed Late Oxfordian age [20], had mixed, predominantly reversed polarity [21–23]. However, its age may be younger than Late Oxfordian (R. O'Sullivan, personal communication, 1984). Furthermore, as a fluviatile sequence, it represents discontinuous sedimentation, and additionally, parts of the magnetic record are poor. The marine clastic sediments of the Swift (Upper Sundance) Formation (western U.S.A.) are Early Oxfordian to latest Callovian and have yielded predominantly reversed polarity, but did not provide a continuous magnetostratigraphic record [22]. Red nodular limestones in Tunisia of Late Callovian and Oxfordian age gave mixed polarity [24], although these sections were discontinuous and lacked biostratigraphic control. The ophiolite complex (gabbros, pillow basalts, and radiolarian cherts) of the Mediterranean island of Elba, mainly Late Oxfordian in age (P.O. Baumgartner, in preparation), has predominantly reversed polarity [25]. However, sampling was too sparse to provide a magnetostratigraphic pattern. Ribbon cherts of Oxfordian and Callovian age in the Lombardy Basin of northern Italy did not give reliable results [26]. Pelagic variegated claystones of Oxfordian age, recovered at DSDP Site 534, yielded mixed polarity and indicated a minimum of 5 reversed zones; however, sparse recovery and non-definitive biostratigraphic data prevented derivation of a polarity sequence [9].

Of these studies which yielded results, all indicated mixed to predominantly reversed polarity during Oxfordian time. In contrast, studies of shallow marine limestones, of Franconia in southern Germany [27], and of the Jura Mountains in Switzerland [28], found entirely normal polarity for the whole Oxfordian and Early Kimmeridgian, with perhaps a few reversed events. In central

Italy, siliceous limestones of the Calcari Diaspringi Formation also yielded entirely normal polarity [29; summarized in 30]. This formation is inferred to be Oxfordian/Callovian in age from regional stratigraphic considerations; the formation itself has no age diagnostic fauna. The results of these three predominantly normal polarity studies are in conflict, both with the marine magnetic anomaly record (Fig. 1), and with the five studies yielding mixed polarity. For several reasons, we believe that the normal polarity studies may be the result of incomplete removal of a normal overprint: (1) the Franconia and Jura results are from shallow marine limestones, a lithology which often yields poor paleomagnetic results; (2) normal is the most frequent polarity of overprinting magnetization; and (3) the disagreement with the mixed polarity of the marine record. In the case of the Calcari Diaspringi Formation (not shallow marine), the evidence for an Oxfordian age of the strata is poor, thus it may not be Oxfordian.

We have sampled Middle and Late Oxfordian pelagic carbonates in northern Spain. These strata have detailed ammonite zonation and have proved to be reliable paleomagnetic recorders. The results have: (1) shown that this portion of the Oxfordian had a moderately high reversal frequency (about five per million years); (2) yielded a polarity pattern that is reproduced in detail among four sections; and (3) provided biostratigraphic ages of the reversals from ammonite zonation and subzonation, enabling construction of a portion of the Oxfordian polarity time scale. Our results indicate a much greater frequency of reversals than do any of the sea-floor models, and may provide an explanation of the dereasing anomaly amplitudes and the "quiet zones".

2. Geologic setting and stratigraphy

A series of outcrops of Callovian and Oxfordian sedimentary rocks are located just north of the village of Aguilon in the central part of the northern (Aragonian) branch of the Iberian Cordillera (Fig. 2). The site (41.3°N, 1.0°W) is approximately 52 km south of Zaragoza. The Oxfordian stratigraphy of the Iberian Cordillera has been well documented (for example, [31,32]). Systematic collecting and study of the rich ammonite fauna has led to a detailed ammonite zonation [33,34]. This zonation is essentially the same as that of the submesogean province of the Tethyan realm [35], with proposed modifications using *Perisphinctid* ammonite faunas [36]. This paper will use the French convention of assigning the *P.(D.) bifurcatus* zone to the Late Oxfordian [35], although some zonations and a recent proposal by the Oxfordian Working Group of the International Subcommission on Jurassic Stratigraphy (in preparation) place this zone into an expanded Middle Oxfordian.

A suite of five overlapping outcrop sections was sampled within a 1- to 2-km^2 area centered about 1 km north of Aguilon. These were denoted Aguilon (AG) 1 through 5 in accordance with the nomenclature of Sequeiros and Melendez [33,34].

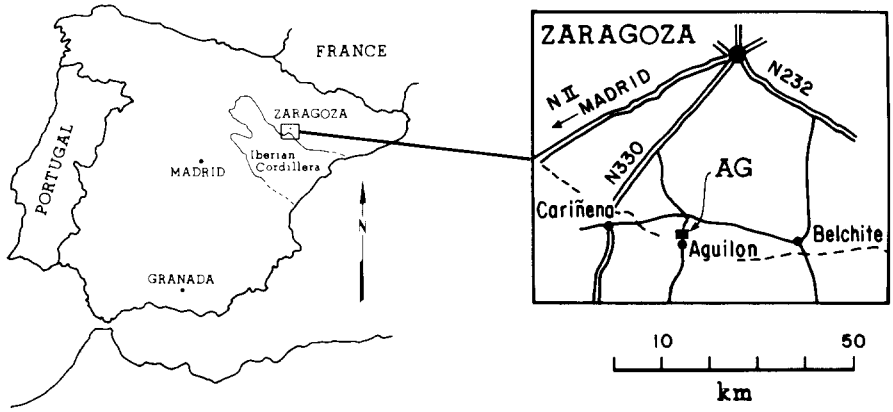


Fig. 2. Location of the magnetostratigraphic sections (*AG*) near Aguilon, at the northern edge of the Iberian Cordillera in Spain.

A composite stratigraphic column is shown in Fig. 3. Paleomagnetic sampling began in the Early and Middle Callovian, 25 m of interbedded grayish yellow marly limestone and marl (section AG-1). This lithology possessed very weak magnetization and displayed a lack of coherent magnetic behavior upon demagnetization, and AG-1 was discarded. The Late Callovian and Early Oxfordian are extremely condensed or absent, being represented by two thin, irregular iron-oolite limestone beds, each capped by a hardground.

Continuous pelagic sedimentation resumed in the late Middle Oxfordian (Fig. 3) and continued into the Late Oxfordian. Biostratigraphically and lithostratigraphically, sedimentation appears to have been continuous except for a possible, biostratigraphically brief, hiatus at the top of the *Lithacoceras (Larcheria) schilli* horizon. The lithology consists of gray, massive, sponge-rich, biomicrite limestone beds (20–50 cm thick) of Middle

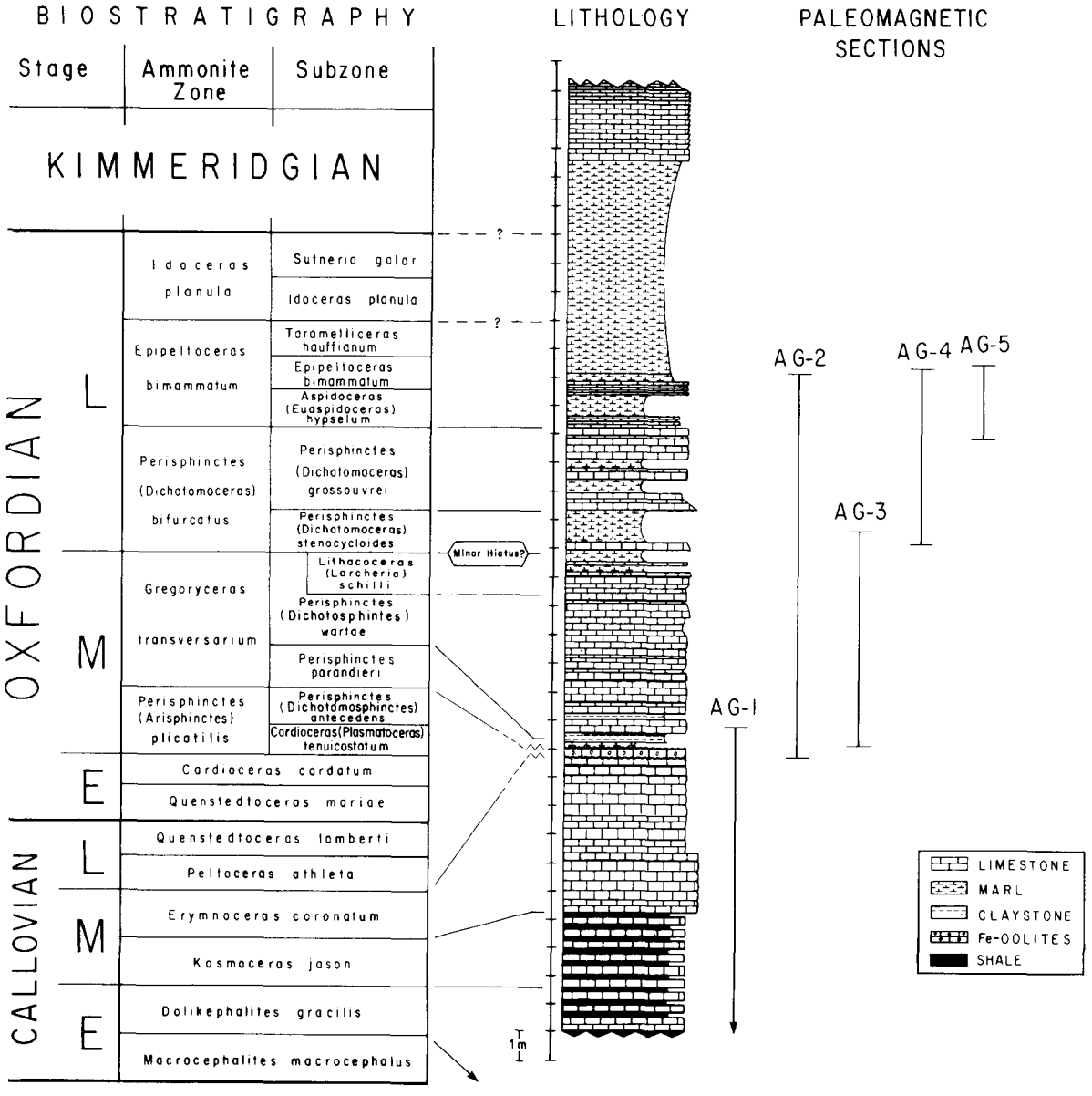


Fig. 3. Lithology and biostratigraphy of the five magnetostratigraphic sections of Aguilon. Meter scale is to left of the lithologic column. Section AG-1 was continued downward into the upper Bathonian.

Oxfordian age overlain by Upper Oxfordian interbedded, gray sponge-bearing limestone (10–25 cm thick beds) and marly intervals (which progressively increase in thickness vertically from a few centimeters to 40–50 cm). Above, marl with sparse, poorly preserved ammonites becomes dominant (upper part of the *E. bimammatum* zone) and was not sampled. The top of this marl appears to be a condensed horizon containing the *Taramelliceras hauffianum* subzone and lower part of the *Idoceras planula* zone. The upper part of the *I. planula* zone and the Oxfordian/Kimmeridgian stage boundary cannot be delimited in this marl unit. The overlying rhythmically-bedded micritic limestones of Kimmeridgian/Tithonian age lack ammonite biostratigraphy.

The placement of the ammonite subzone boundaries is determined by the lowest occurrence of the index fossil; therefore with more extensive paleontological sampling the actual boundary may be found to be slightly lower than shown in Fig. 3, but not higher. Paleontological studies were generally on a bed-by-bed scale, therefore there is an inherent 20–40 cm uncertainty for any boundary. Other biostratigraphic uncertainties, caused by poorly-defined ammonite assemblages, are shown on the stratigraphic and magnetostratigraphic columns by a shaded band of appropriate width.

Sections AG-2, -3 and -5 are on the south limb of a broad anticline and have average dips/azimuths of 48°/161°, 40°/177°, and 19°/188° respectively. Section AG-4 (1 km north of AG-2) is on the north limb of the anticline; within the AG-4 outcrop the beds display a small scale fold, attitudes ranging from 18°/200° to 37°/10°. Thus, section AG-4 contains a fold test, and the four sections together provide a second fold test. The age of this deformation is probably Oligocene or Early Miocene [37].

3. Technique

The density of field sampling was determined by the estimated average rate of sedimentation and expected frequency of magnetic reversals. A sampling program of 20 samples per ammonite zone (or an estimated interval of 50,000 years between samples) proved to be too sparse for the high frequency of reversals encountered, so the sampling density was doubled in the second field season. A total of 105 samples were cored from the 13 m represented in four sections (AG-2 through 5). Samples were trimmed to 2.4-cm cylinders.

Measurements were made on a two-axis ScT cryogenic magnetometer (noise level 2×10^{-11} A/m) with on-line computer data analysis. The magnetometer and the non-inductively wound demagnetization furnace with separate cooling chamber are contained within a steel shielded room (internal field less than 1000 nT). Each sample was measured in eight orientations (90° rotations on the z-axis in the upright and inverted positions), and the measurements averaged.

Two dozen pilot samples representative of the various lithologies underwent progressive thermal demagnetization in 40–50°C increments from 150°C to 650°C. Based on the pilots, the remaining samples were measured at a selected set of about eight demagnetization steps. Magnetic polarity and characteristic direction of remanent magnetization were determined for each sample individually. A three-dimensional least-squares line fit technique [38] was used to determine the mean characteristic direction of magnetization and associated variance for each sample. The set of vectors (direction and intensity of magnetization) used for this analysis were generally those encompassing the 400°C through 525°C demagnetization steps, but varied from sample to sample depending upon its temperature of "cleaning" and stability as judged from vector plots of the demagnetization path. The mean characteristic direction derived from this least-squares technique is a more accurate and less biased representation of the stable magnetization than selecting either a single uniform temperature step for all, or a "best" step for each, of the samples [38].

4. Remanent magnetization

The natural remanent magnetization (NRM) directions of the samples at each site are shown in Fig. 4. The NRM directions are not that of the axial or present earth's field (000.0, 60.4 and 355.0, 56.0) at any of the four sites; instead, the directions are intermediate between the present field direction and the characteristic direction after thermal demagnetization. NRM intensities are approximately 2.5×10^{-3} A/m for three of the sites;

the fourth (AG-4) has higher NRM intensities, averaging 3.8×10^{-3} A/m.

Thermal demagnetization was reasonably effective in removing secondary magnetizations. In general all samples were demagnetized at 50°C steps, from 250°C through 500°C, then at 525°C and 550°C. The samples of all sites responded similarly to demagnetization, as would be expected from the uniform facies. Intensities drop sharply; at 350°C they are 5–20% of the NRM intensities (Fig. 5). Directional change begins at or above this temperature; in general by 500°C, most sites display clustering of normal and reversed directions (Fig. 6). The loss of nearly all of the magnetization by the 550°C step suggests that the principal carrier of the characteristic magnetization is magnetite.

5. Mean directions

The results of the between-sites and within-bed fold tests are shown in Table 1. Structural corrections yield a better clustering (higher estimate of precision parameter, k) of the Aguilon directions

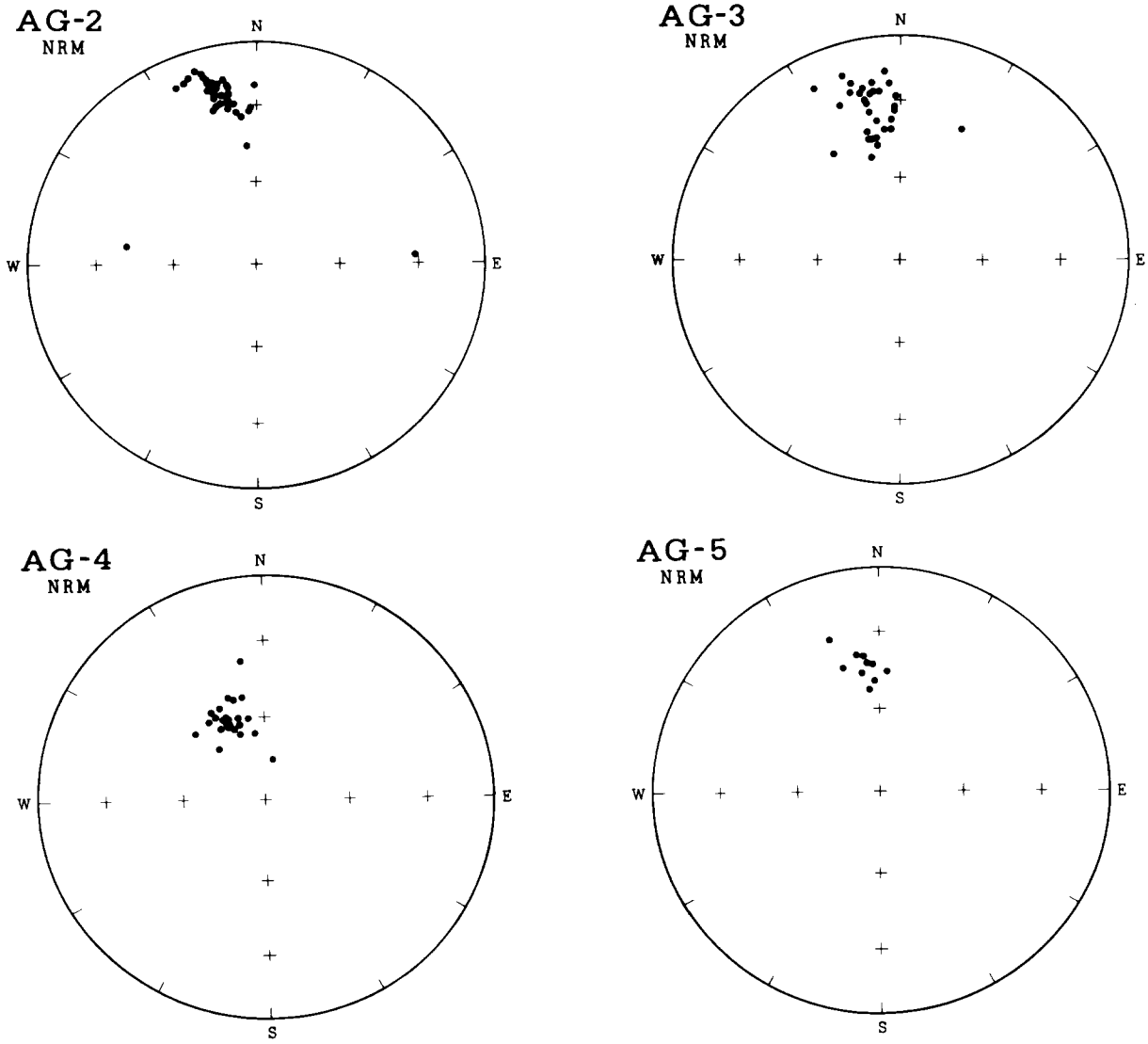


Fig. 4. Natural remanent magnetization (NRM) directions of the Augilon sections AG-2 through AG-5. In all of the equal area plots, solid symbols are lower hemisphere, open symbols are upper hemisphere. Directions are uncorrected for dip of the strata.

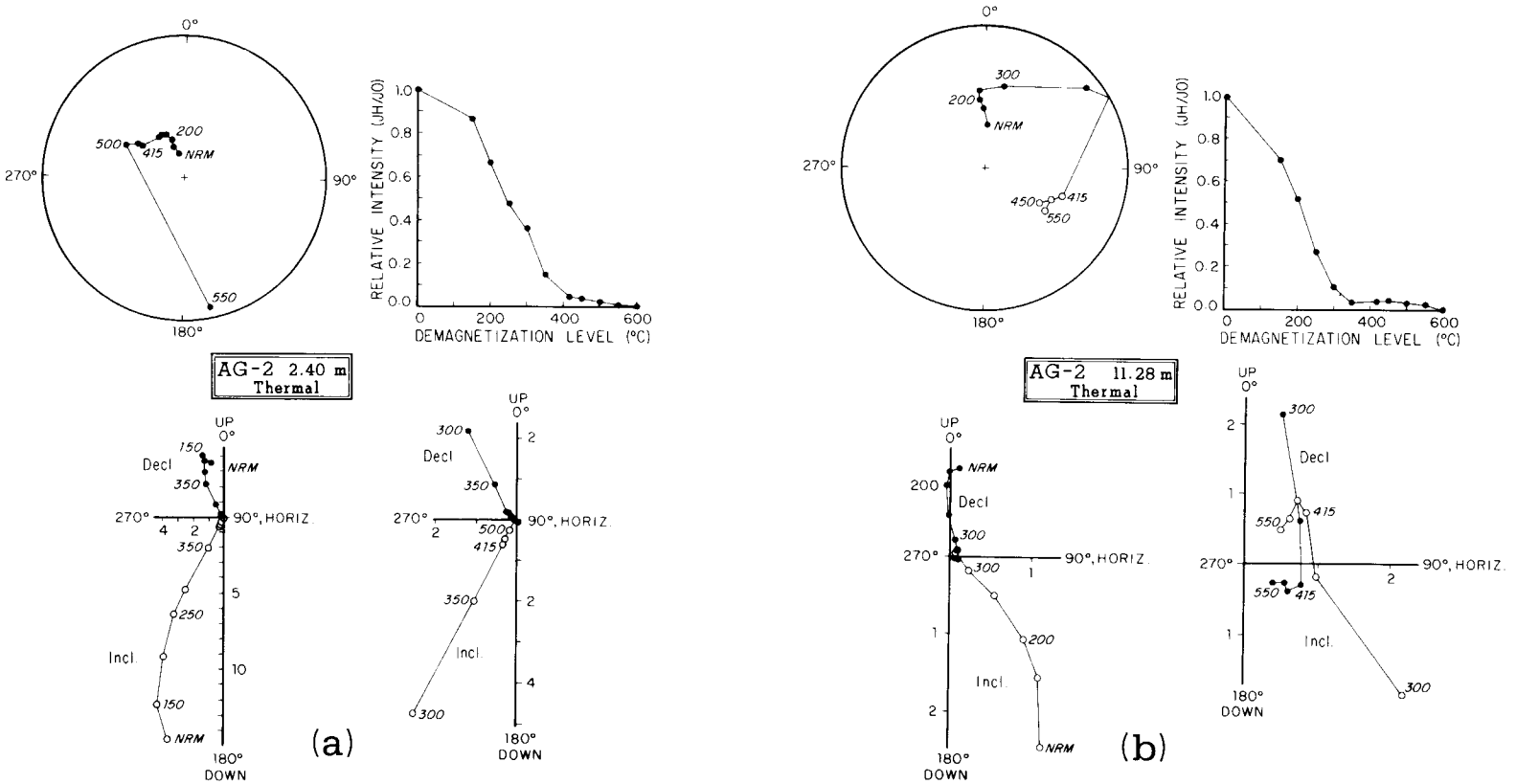


Fig. 5. Thermal demagnetization behavior of two samples of typical gray limestones from sections AG-2. (a) Displays a typical normal sample, and (b) a typical reversed one. Upper left diagram in each set is an equal area plot of the directions during demagnetization (solid and open symbols as in Fig. 4). Upper right diagram in each set displays normalized intensity versus temperature. The lower left diagram in each set is a vector demagnetization diagram of the form used by Kirschvink [38], which enables direct measurement of the stable characteristic magnetizations: declination (solid symbols) is the horizontal component of the demagnetization path; inclination (open symbols) plots the angle from horizontal with the vector length equal to the total intensity at the given demagnetization step; one scale division is equal to 10^{-4} A/m (10^{-7} emu/cm^3) in (a) and 10^{-3} A/m (10^{-6} emu/cm^3) in (b). The lower right diagram in each set is an enlargement of the vector demagnetization diagram to show the behavior at higher-temperature demagnetization steps; one scale division is equal to 10^{-4} A/m (10^{-7} emu/cm^3) in both sets. Steps are in °C; the 600°C and 650°C steps have been omitted because the intensities dropped to the noise level of the magnetometer. Directions in all plots are corrected for tilt of bedding. The odd direction in (a) at 550°C is spurious, caused by the low signal to noise ratio at this demagnetization step (intensity of magnetization = 5×10^{-6} A/m).

as a group, and an improvement in the antipodal nature of the normal and reversed groups, thus demonstrating that the characteristic magnetization was set prior to the regional folding.

Mean directions of characteristic magnetization and Fisherian statistics were calculated both with and without points lying two standard deviations away from the mean. Mean directions were computed for both the characteristic directions of the normal and reversed polarity groups independently, and for all samples combined; Table 1 shows these results. The normal and reversed means are not antipodal, suggesting the presence of a persistent overprint. The average intensity of magnetization of the mean characteristic direction of normal polarity is greater than that of the reversed polarity at all sites, indicating that the overprint is relatively closer in direction to the mean normal direction. This overprint clearly biases the reversed directions substantially. In an

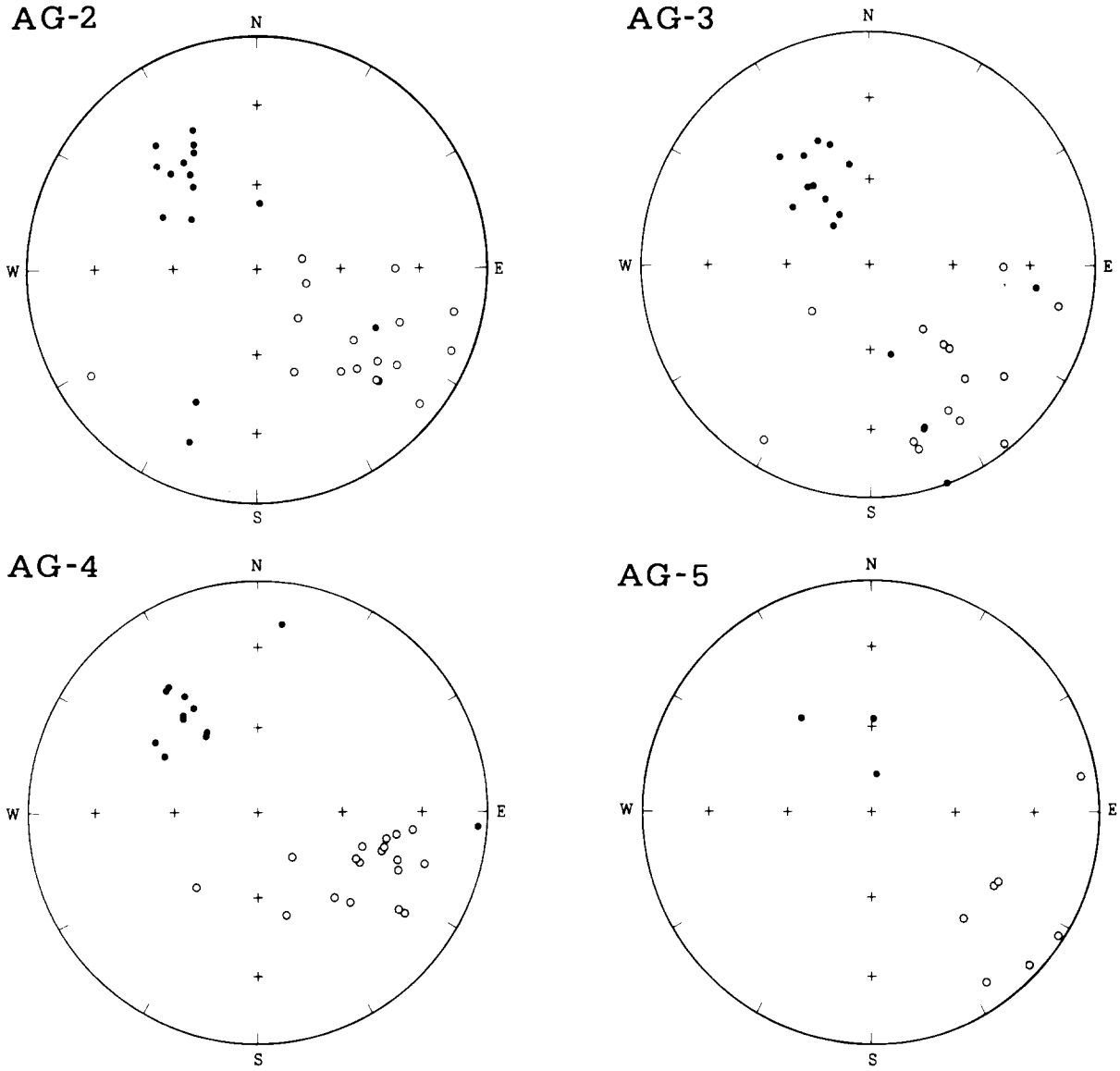
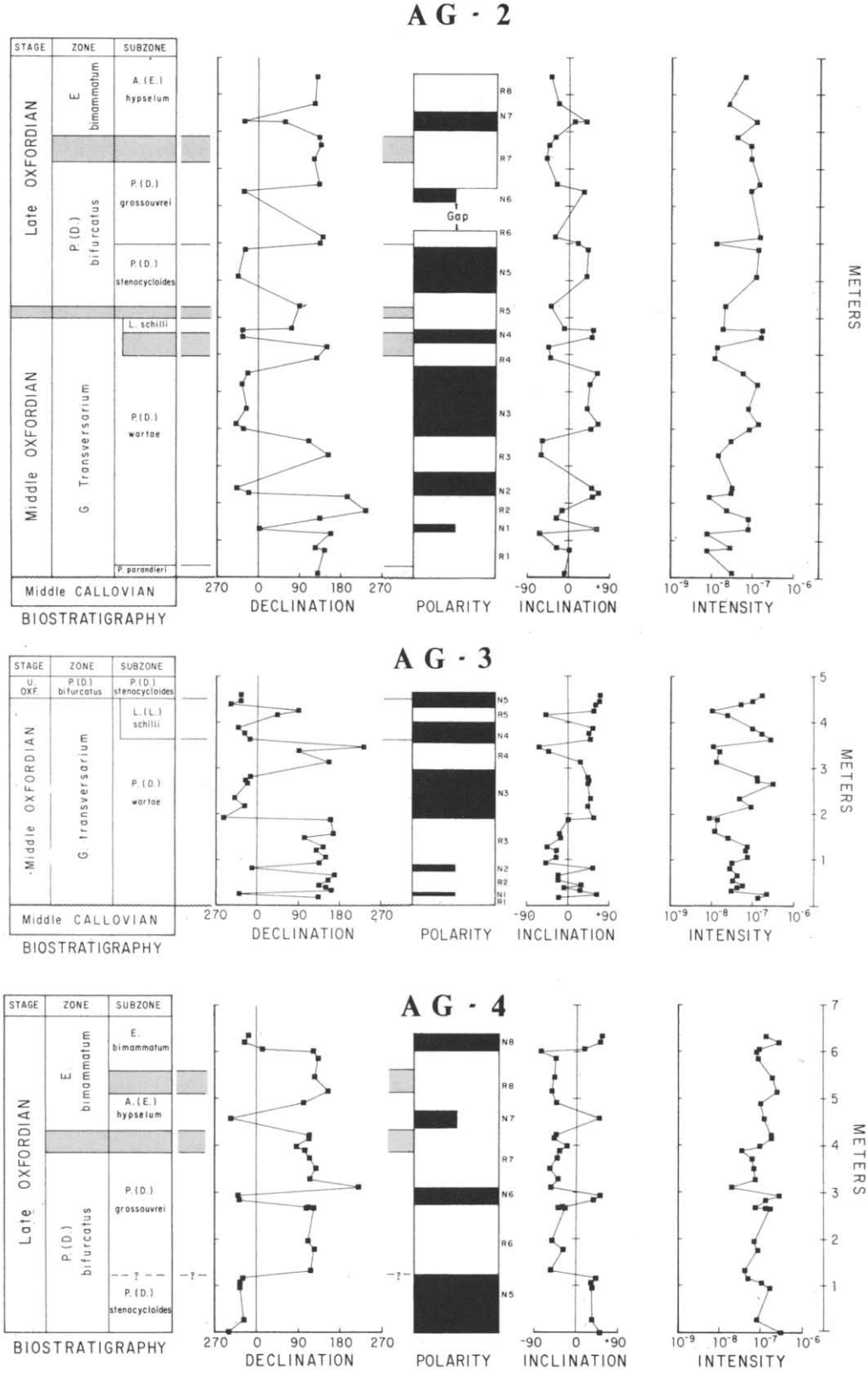


Fig. 6. Characteristic directions of magnetization for the Aguilon sections, as derived from a least-squares line fit of the vectors at generally the 450°C, 500°C, and 525°C demagnetization steps (see text). Equal area plots (symbols as in Fig. 4); directions are corrected for dip of the strata.

TABLE 1

Mean directions

Section	Samples	N/Total	Mean characteristic direction					VGP			Paleo-latitude (°N)
			D (°)	I (°)	α_{95} (°)	K	Intensity (10^{-5} A/m)	Lat. °N	Long. °E	(dp, dm)	
AG-2	normal	14/36	328.0	49.8	6.4	39	2.1	62.2	256.7	5.7, 8.6	30.6
	reversed	16/36	313.6	36.4	9.3	17	0.9	45.6	255.2	6.3, 10.8	20.2
	combined	30/36	319.7	43.0	6.2	19	1.5	52.9	255.7	4.8, 7.7	25.0
	weighted	30/36	323.7	45.8			1.5	57.2	255.2		27.2
AG-3	normal	15/34	330.8	54.3	7.0	31	2.7	66.3	262.7	6.9, 9.9	34.8
	reversed	13/34	320.5	16.3	18.4	6	0.7	42.0	236.8	9.8, 19.0	8.3
	combined	28/34	325.1	38.1	11.1	7	1.8	54.6	245.7	7.8, 13.1	21.4
	weighted	28/34	328.7	46.5			1.7	61.1	251.1		27.8
AG-4	normal	10/30	325.0	44.5	7.4	44	3.0	57.5	252.4	5.8, 9.3	26.2
	reversed	18/30	296.1	43.8	7.4	23	2.2	35.7	273.5	5.8, 9.2	25.6
	combined	28/30	306.4	44.9	6.3	20	2.5	43.9	267.8	4.9, 7.9	26.5
	weighted	28/30	312.8	44.2			2.6	48.4	262.5		25.9
AG-5	normal	3/10	341.3	62.5	19.9	39	0.5	76.0	254.3	24.3, 31.1	43.8
	reversed	3/10	304.8	20.3	20.9	9	0.5	32.8	252.8	11.4, 21.8	10.4
	combined	10/10	311.4	34.2	20.8	6	0.5	43.0	255.4	13.6, 23.8	18.7
	weighted	10/10	323.0	41.4			0.5	54.6	251.0		23.8
Fold tests											
All samples	geogr.	110/110	318.0	17.2	7.1	4	1.8				
	str.	110/110	316.3	43.0	5.4	7	1.8	50.5	258.6	4.2, 6.7	25.0
AG-4 single bed	geogr.	3/3	285.0	21.1	23.8	28	12.7				
	str.	3/3	291.0	20.7	8.0	238	12.7	22.8	203.2		
All samples	combined	96/110	315.8	40.1	5.4	8	1.6	48.2	255.4	4.2, 6.7	22.8
	weighted	96/110	322.1	44.5	3.1	22	1.6	55.2	251.8	2.4, 3.8	25.6
All sites	combined	4/4	315.8	40.3	9.1	103	1.6	48.8	256.5	6.6, 11.0	23.0
	weighted	4/4	322.0	44.6	6.0	239	1.6	55.4	255.4	4.7, 7.5	26.3


AG - 2
STAGE
ZONE
SUBZONE
Late OXFORDIAN
E. bimammatum
A. (E.) hypselum
P. (D.) bifurcatus
P. (D.) grossouvrei
P. (D.) stenocycloides
Middle OXFORDIAN
G. Transversarium
L. schilli
P. (D.) wartae
P. parandieri
Middle CALLOVIAN
BIOSTRATIGRAPHY
Gap
R8
N7
R7
N6
R6
N5
R5
N4
R4
N3
R3
N2
R2
N1
R1
270 0 90 180 270
DECLINATION
POLARITY
-90 0 +90
INCLINATION
10^{-9} 10^{-8} 10^{-7} 10^{-6}
INTENSITY
METERS
AG - 3
U. OXF.
P. (D.) bifurcatus
P. (D.) stenocycloides
L. (L.) schilli
G. transversarium
P. (D.) wartae
Middle OXFORDIAN
Middle CALLOVIAN
BIOSTRATIGRAPHY
N5
R5
N4
R4
N3
R3
N2
R2
N1
R1
270 0 90 180 270
DECLINATION
POLARITY
-90 0 +90
INCLINATION
10^{-9} 10^{-8} 10^{-7} 10^{-6}
INTENSITY
METERS
1 2 3 4 5
AG - 4
E. bimammatum
A. (E.) hypselum
P. (D.) bifurcatus
P. (D.) grossouvrei
P. (D.) stenocycloides
Late OXFORDIAN
BIOSTRATIGRAPHY
N8
R8
N7
R7
N6
R6
N5
270 0 90 180 270
DECLINATION
POLARITY
-90 0 +90
INCLINATION
10^{-9} 10^{-8} 10^{-7} 10^{-6}
INTENSITY
METERS
1 2 3 4 5 6 7

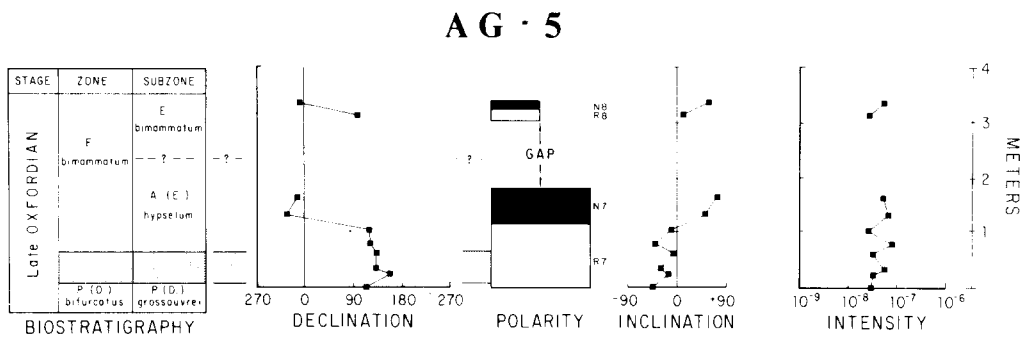


Fig. 7. Characteristic directions of magnetization (declination and inclination) plotted in stratigraphic sequence, for all four sections. The directions shown are those computed by least-squares line fit of the vectors at generally the 450°C, 500°C and 525°C demagnetization steps (see text); the plotted intensity of magnetization is the mean intensity computed for these vectors (the intensity scale is in emu/cm^3; 1 emu/cm^3 = 10^{-3} A/m). Black is normal polarity. Single sample polarity intervals are represented by half bars. The nomenclature for polarity intervals is based on the composite scale (Fig. 9). Maximum uncertainties in the placement of ammonite subzone boundaries are indicated by the width of the stippled shading.

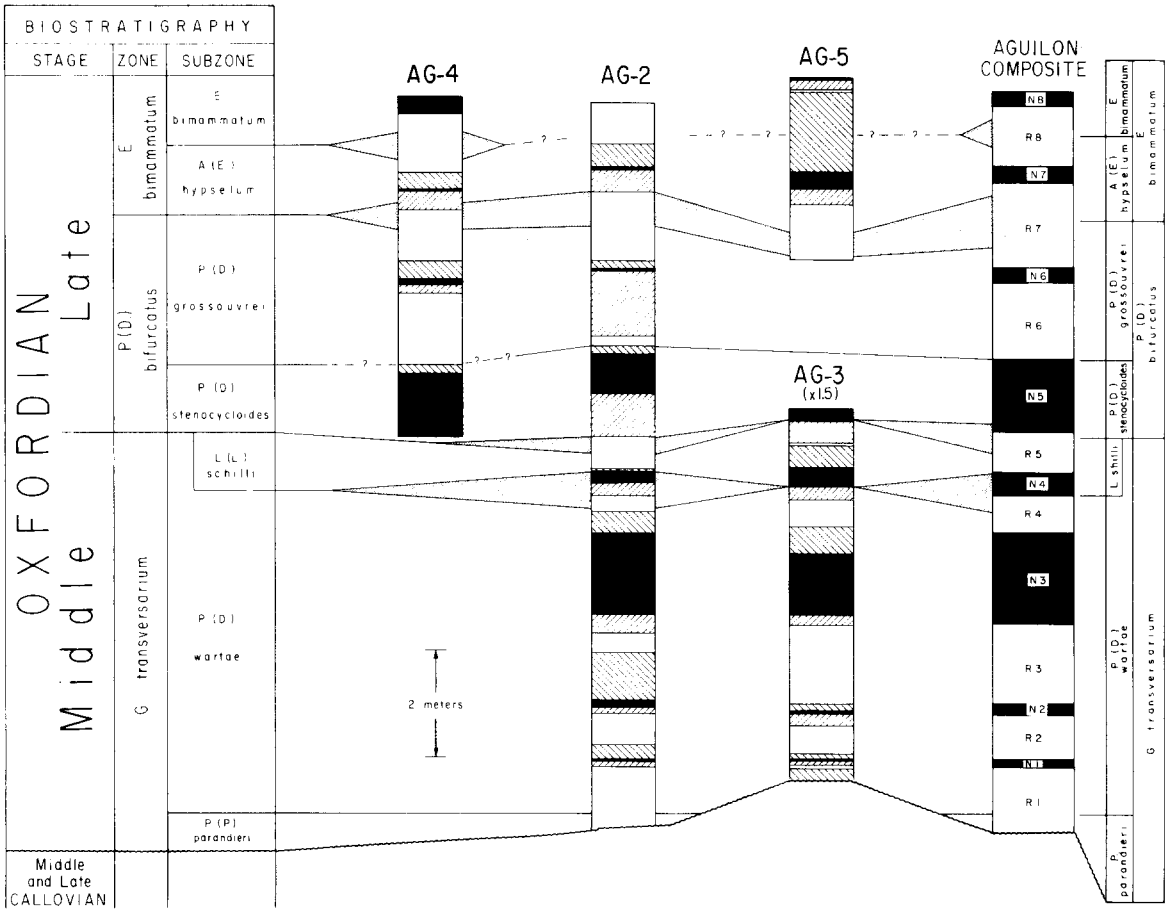


Fig. 8. Correlation of the polarity interpretations for the Aguilon sections based on the ammonite zonations. Maximum uncertainty in the placement of ammonite zone boundaries (see text) is indicated by the stippled shading. Cross-hachured intervals in the polarity columns indicate the total stratigraphic distance between samples of opposite polarity; this is the maximum uncertainty in the placement of the magnetic polarity interval boundaries. The composite polarity scale at the far right is constrained by optimizing the width of each polarity zone in such a way that it would not conflict with the different polarity columns in which it is represented. For this diagram, section AG-3 was linearly expanded by a factor of 1.5 (see text). Polarity intervals in the composite are numbered from the base.

attempt to compensate for the effect of this overprint, weighted means were computed, using the average intensities of magnetization of the normal and reversed directions as a weighting factor. These mean directions are also shown in Table 1. We then computed an overall mean direction, both from the unweighted and weighted mean directions, treating each section as a single site. The weighted mean direction is 322.0°, 44.6° (α_{95} = 6.0°), which is only slightly different from the unweighted calculation, but has a significantly higher precision.

6. Magnetostratigraphy

The stratigraphic distribution of characteristic magnetic directions and the associated polarity interpretation of each of the the Aguilon sections is shown in Fig. 7. The uncertainties associated with the placement of ammonite zone boundaries are indicated by stippled shading. Cross-hachuring denotes the maximum uncertainties in the thicknesses of the polarity zones that result from the spacing of the samples.

Fig. 8 compares the magnetostratigraphic and biostratigraphic zonations of all four sections. Site AG-3 has been linearly expanded in Fig. 8 by a factor of 1.5 relative to the scale of the other sites. This was done because in its original form, the ammonite subzones of AG-3 were much reduced in thickness relative to the other sections. The expansion facilitates the visual comparison of the magnetostratigraphic zones within each ammonite subzone, between sections AG-2 and AG-3. The factor of 1.5 was chosen to make the apparent width of the *G. transversarium* ammonite zone of AG-3 similar to that of section AG-2; this also seemed to produce a good match between the magnetostratigraphic zonation, as well as the ammonite zonation, of the two sections. The implied slower sedimentation rate at AG-3 may be a function of an irregularly eroded Middle-Late Callovian depositional surface, consistent with other geologic indications of a long cessation of deposition and extensive erosion and/or dissolution.

The correlation of polarity among the sites, using the ammonite zonation for control, is quite good. In general, polarity intervals are represented by two or more samples. Because of the high frequency of reversals, there are one to three polarity intervals at each site which are represented by only one sample. However, when all sites are viewed together, tied by their common ammonite zonation, polarity intervals represented by a single sample in one site are documented by multiple samples in another site. There is one exception; the lowest normal polarity interval (N1 in the *P.(D.) wartae* subzone, Fig. 8) is represented by single samples in both of the two sections (AG-2, AG-3) encompassing that time. Since nearly all intervals are represented by multiple samples, it would appear that the continuity of sedimentation and sampling density at each site was adequate to record and identify each polarity interval, albeit minimally in a few places.

A composite magnetic polarity column was constructed by taking the average documented thickness of each polarity interval among the four sections, plus half of the residual uncertainty in the precise placement of the bounding reversals. The resultant composite polarity column in Fig. 8 is fairly well defined by the duplication among the sections. Slight distortions may exist in the composite pattern, relative to the true geomagnetic history, as a result of sedimentation processes; however, these distortions are small and do not greatly affect the general validity of the geomagnetic polarity pattern obtained. One distortion may result from the possible, although not certain, biostratigraphically brief hiatus at the top of the *L.(L.) shilli* ammonite horizon; this may contribute a minor distortion to the widths of the R5 and N5 polarity intervals. The second, and more significant source of distortion, is the decrease in carbonate content in the Late Oxfordian portion of this sequence. This may have caused the polarity intervals N5 through N8 to appear compressed with respect to the lower intervals. This change in lithology is the only indication of a change in sedimentation rate, and there are no other indications of a lack of continuity in sedimentation or in rate through the time interval represented here. Thus, this composite probably represents a good approximation of the magnetic polarity history for this part of the Oxfordian.

7. Estimated ages of pre-M25 marine magnetic anomalies

In order to compare our results with the marine pattern, it was necessary to estimate the biostrati-

graphic ages of the Pacific and Atlantic magnetic anomaly block models using recent absolute time scales for the Jurassic. Predicted ages for the two block models were generated using two different recent time scales, that of Kent and Gradstein [39,40] and that of Westermann [42,43]. The two block models with the Westermann time scale ages were shown in Fig. 1. The ages of the block models were derived separately. For the Pacific model of Cande et al. [3], the ages were derived by assuming a constant spreading rate with age control points of: (1) the Tithonian/Berriasian boundary = lower boundary of M18, a compromise between the various paleontological definitions of this stage boundary [9,13], and = 144 Ma [39–43]; and (2) the Oxfordian/Kimmeridgian boundary = upper boundary of M25 [14], and = 156 Ma in the Kent-Gradstein time scale [39,40] or = 156.8 Ma in the Westermann time scale [42,43]. For the Atlantic model of Bryan et al. [18], the same age was used for M25 and a Middle Callovian age was assigned to the upper boundary of their "M28", based on the results of DSDP Site 534 [44]; the resulting spreading rates for the Blake-Bahama Basin, from which the magnetic anomaly survey was derived, were 31.5 km/Ma and 34.4 km/Ma in the Kent-Gradstein and Westermann time scales respectively.

8. Correlation of Aguilon to the M sequence

The composite magnetostratigraphy of the Aguilon sections spans the uppermost quarter of the Middle Oxfordian and the lower half of the Late Oxfordian, or a total of about 4.5 ammonite subzones. In Fig. 9, the Aguilon composite has been scaled to 4.5 ammonite subzones (approximately 1.5 m.y.) on the Westermann "equal-subzone" time scale [42,43]; the composite is positioned opposite the corresponding age interval of the Pacific pre-M25 block model, using the ages derived in the preceding section. There has been no attempt to correct the Aguilon composite for the unequal widths of its component subzones. As noted, in general, the younger subzones have a reduced thickness relative to the older ones. This reduction in relative thickness of ammonite subzones also suggests an upward decrease in sedimentation rate, consistent with the same inference derived from the observed decrease in carbonate content.

The Aguilon polarity composite is not the same pattern as Cande et al.'s [3] block model of the Pacific anomalies. The decrease in sedimentation rate with time at Aguilon may contribute to the mismatch, but a more major factor is probably the rate of reversals combined with the characteristics of the two different recording mechanisms.

Many factors go into the correct interpretation of a marine magnetic anomaly sequence. When the signal intensity is as low as it is in the Jurassic "quiet zone", a great many more factors become critical in deriving the correct geomagnetic polarity history from such a record than is the case in most marine anomaly records. Some possible sources of error are: (1) reversal rate relative to spreading rate, and both in relation to the sea-floor intrusion mechanism of magnetic recording; (2) the necessary assumption of a constancy of spreading rate, in the face of a lack of information regarding rate of spreading: virtually no good spreading rate information is available for this time; (3) skewness or the location of the oceanic crust when generated; (4) location of the crust when the magnetometer survey is made: anomalies recorded in areas where diurnal variations are large (as some of these records were), will have significant pertubations in such a low amplitude signal; (5) type of survey: deep- or mid-level tow is needed to resolve details in such low amplitude records; (6) regional field correction becomes critical for low-amplitude records; (7) age control is very weak for the old end of the marine anomaly record.

If the reversal rate is as high as is indicated by the Aguilon sequence, the sedimentary record at Aguilon is likely to be a more faithful reproduction of the details of the geomagnetic polarity history of this time than are the surface-tow marine records. Several of the brief polarity intervals displayed by the Aguilon sequence are probably less than 100,000 years in duration; with a half-spreading rate of 3 cm/yr, such episodes would be equivalent to blocks of crust only 3 km in width. That width is less than the usual width of transition zones between crustal blocks of opposite polarity.

The very small anomaly amplitudes of this time probably contribute to the mismatch between the marine and the sedimentary records, in the relative proportions of normal and reversed polarity. At

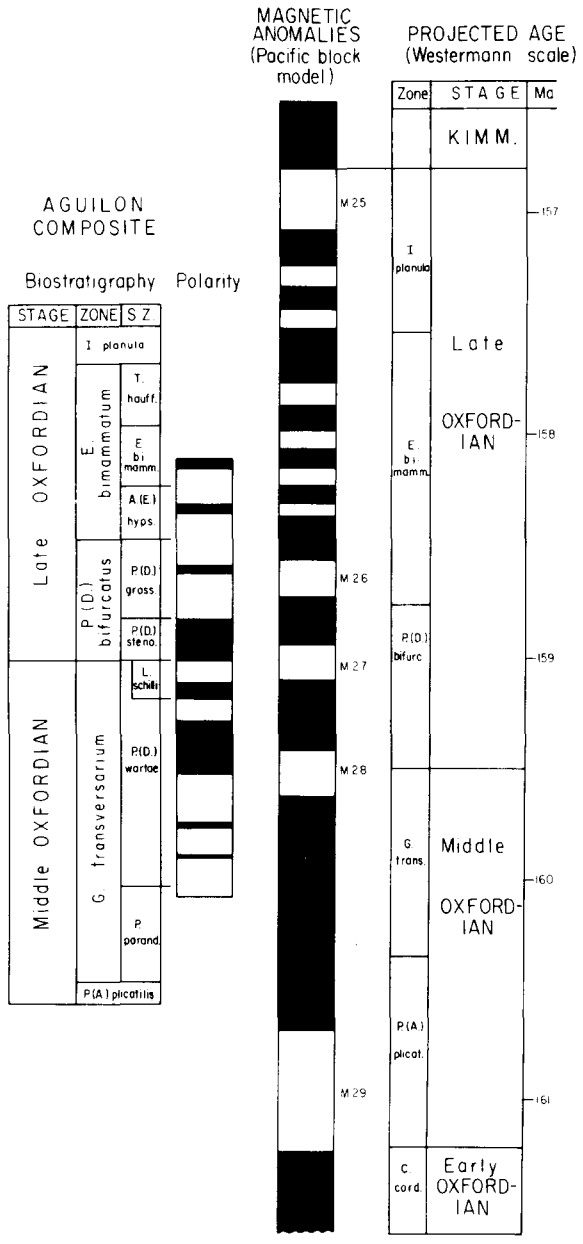


Fig. 9. Comparison of the composite magnetic polarity pattern of the Aguilon sections with the appropriate time interval of the Pacific block model of Cande et al. [3], using ages determined from the Westermann time scale [47,48].

least two factors in the interpretation of the marine anomalies have a significant bearing on this. First, distinguishing between intensity variations and true reversals of the geomagnetic field in such a small amplitude record is difficult, and somewhat tenuous ([45]; s.c. Cande, personal communication, 1984). Furthermore, with such small amplitudes, if the regional field is not known with precision, the zero-baseline could easily be in error, and hence would affect the relative proportions of normal and reversed polarity that are inferred from the record. We also note that the shape of anomaly M27 is not a single peak, but on most profiles ([16]; D. Handschumacher, unpublished compilation, 1984) it appears to have a shoulder; therefore, M27 may be subdivided by a short normal polarity interval.

The ages of the marine anomalies are an approximation, at best. Furthermore, without much information on spreading rate or spreading rate constancy, it is possible that the Aguilon record is correlative with some other part of the marine record. The slightly older portion of the record (M30–M34 [19]) resembles the Aguilon pattern.

9. Pole position and rotation of site

The pole position computed from the weighted mean direction is 255.4°E, 55.4°N (dp, $dm = 2.4$, 3.8). This mean pole from the Aguilon Oxfordian strata located in the central Iberian Cordillera is significantly different from the Kimmeridgian pole position (203.7°E, 70.4°N; dp, $dm = 4.4$, 7.3) that was derived from a site 130 km to the northwest near Soria, on the edge of the Iberian Cordillera [46]. However, the paleolatitudes for both the Aguilon and Soria sites are identical: 26°N (± 2). Furthermore, an Early to Middle Jurassic pole for stable Iberia [47] of 236.2°E, 70.9°N ($\alpha_{95} = 7.3°$) also yields a paleolatitude similar (29°N) to that obtained at Aguilon.

The Mesozoic paleomagnetic poles of Africa form a tight group, and Schott et al.'s [47] pole for stable Iberia falls in the middle of that group. Their result strongly suggests that Iberia moved with the African plate; this result is consistent with the notable similarity of African and Iberian geology and geologic structures. If the Iberian plate was a part of Africa, then there was little or no apparent polar wander with respect to Iberia during Jurassic, and one would expect the three Jurassic pole positions to coincide. The fact that they do not, but that they all give the same paleolatitude, leads us to believe that differential rotation of the thrust sheet containing Aguilon and the block containing Soria during the deformation of

the Iberian Cordillera is the reason for the discrepancies. The rotations required to displace the Aguilon and Soria poles from that of stable Iberia are consistent with the motion expected from the proposed left-lateral shear affecting the northern edge of the Iberian Cordillera in the Aguilon region [48]. The Aguilon pole position suggests a counterclockwise rotation of the thrust sheet containing the sites of about $18 \pm 5°$ relative to stable Iberia.

10. Summary

The magnetic polarity sequence for the Late Middle and early Late Oxfordian has been determined in pelagic limestones having detailed ammonite zonation. The reproducibility of the pattern has been demonstrated by four overlapping magnetostratigraphic sections. The frequency of reversals averages about 5–6 reversals per ammonite zone. The average ratio of reversed polarity to normal polarity is approximately 2.5 : 1.

Using revised estimate of the ages for the block models of the marine magnetic anomalies, the Aguilon polarity sequence probably is equivalent in age to "M26" of the Atlantic model of Bryan et al. [18], and to Cande et al.'s [3] anomalies "M26", "M27", "M28" and some of the polarity structure preceding them. The Aguilon polarity sequence agrees in principle with Cande et al.'s [3] anomaly model; both indicate frequent reversals of the geomagnetic field. However, Aguilon suggests that many more reversals of the field occurred, and also indicates a bias of the field towards reversed polarity. The Aguilon record suggests that a high reversal frequency in the Middle and Late Oxfordian is the explanation for the marine "quiet zones" of this time. This concept and the higher reversal frequency are in agreement with the same conclusions reached by Handschumacher (personal communication, 1985) from his study of the Pacific record.

The mean direction at Aguilon is 322.0°, 44.6°, $\alpha_{95} = 6.0°$. This yields a mean pole position at 255.4°E, 55.4°N. This direction indicates a counterclockwise rotation relative to stable Iberia, presumably caused by left-lateral shear during the Tertiary.

The establishment of this magnetic polarity sequence in well-dated sequences of the Tethyan faunal realm will enable future correlation to the northwest European and Boreal faunal realms. When this is accomplished, it will be possible to correlate geological events during the Oxfordian between different climatic regions of the earth.

Acknowledgements

This magnetostratigraphic research is part of a larger study funded by NSF grant EAR-8107739. Additional support was provided by NSF grant OCE-8207771. We express our appreciation to the government of Spain for giving us permission to work in their country, and to Nancy Winslow and Beverly DeVore, our research assistants, for helping with the measurements and early interpretations. An early manuscript version was reviewed by Drs. L. Tauxe and E.L. Winterer.

References

1 R.L. Larson and W.C. Pitman III, World-wide correlation of Mesozoic magnetic anomalies and its implications, Geol. Soc. Am. Bull. 83, 3645–3662, 1972.
2 R.L. Larson and T.W.C. Hilde, A revised time scale of magnetic reversals for the Early Cretaceous, Late Jurassic, J. Geophys. Res. 80, 2586–2594, 1975.
3 S.C. Cande, R.L. Larson and J.L. LaBrecque, Magnetic lineations in the Pacific Jurassic quiet zone, Earth Planet. Sci. Lett. 41, 434–441, 1978.
4 J.R. Heirtzler, G.O. Dickson, E.M. Herron, W.C. Pitman III and X. Le Pichon, Marine magnetic anomalies, geomagnetic field reversals, and motions for the ocean floor and continents, J. Geophys. Res. 73, 2119–2136, 1968.
5 W. Lowrie and W. Alvarez, One hundred million years of geomagnetic polarity history, Geology 9, 392–397, 1981.
6 J.E.T. Channel, W. Lowrie and R. Medizza, Middle and Early Cretaceous magnetic stratrigraphy from the Cismon section, northern Italy, Earth Planet. Sci. Lett. 42, 153–166, 1979.
7 W. Lowrie, W. Alvarez, I. Premoli-Silva and S. Monechi, Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks, Geophys. J. R. Astron. Soc. 60, 263–281, 1980.
8 J.G. Ogg, Sedimentology and paleomagnetism of Jurassic pelagic limestones: "Ammonitico Rosso" facies, 212 pp., Ph.D. Thesis, Scripps Institution of Oceanography, UCSD, 1981.
9 J.G. Ogg, Magnetostratigraphy of Upper Jurassic and lowest Cretaceous sediments, Deep Sea Drilling Project Site 534, western North Atlantic, Init. Rep. DSDP 76, 685–697, 1983.
10 B. Galbrun and L. Rasplus, Magnetostratigraphie du stratotype du Berriasien: Premiers resultats, C.R. Acad. Sci. Paris, Ser. II 298, 219–222, 1984.

11 W. Lowrie and J.E.T. Channell, Magnetostratigraphy of the Jurassic-Cretaceous boundary in the Maiolica limestone (Umbria, Italy), Geology 12, 44–47, 1984.

12 J.G. Ogg, Comment on "Magnetostratigraphy of the Jurassic-Cretaceous boundary in the Maiolica limestone (Umbria, Italy)", Geology, in press, 1985.

13 S. Cirilli, P. Marton and L. Vigliotti, Implications of a combined biostratigraphic and palaeomagnetic study of the Umbrian Maiolica formation, Earth Planet. Sci. Lett. 69, 203–214, 1984.

14 J.G. Ogg, M.B. Steiner, F. Oloriz, and J.M. Tavera, Jurassic magnetostratigraphy, 1. Kimmeridgian-Tithonian of Sierra Gorda and Carcabuey, southern Spain, Earth Planet. Sci. Lett. 71, 147–162, 1984.

15 E. Irving and G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, Earth Sci. Rev. 12, 35–64, 1976.

16 D. Handschumacher and L. Kroenke, An extended time scale of Mesozoic geomagnetic reversals, EOS 59, 266, 1978.

17 D.L. Barrett and C.E. Keen, Mesozoic magnetic lineations, the magnetic quiet zone, and sea-floor spreading in the Northwest Atlantic, J. Geophys. Res. 81, 4875–4884, 1976.

18 G.M. Bryan, R.G. Markl and R.E. Sheridan, IPOD site surveys in the Blake-Bahama Basin, Mar. Geol. 35, 43–63, 1980.

19 D. Handschumacher and J.S. Gettrust, Marine Geophysics at NORDA: Sea Floor Investigations, Naval Res. Rev., in press, 1985.

20 R.W. Imlay, Jurassic paleobiogeography of the conterminous United States in its continental setting, U.S. Geol. Surv. Prof. Pap. 1062, 134 pp., 1980.

21 M.B. Steiner and C.E. Helsley, Reversal pattern and apparent polar wander for the Late Jurassic, Geol. Soc. Am. Bull. 86, 1537–1543, 1975.

22 M.B. Steiner and C.E. Helsley, Late Jurassic magnetic polarity sequence, Earth Planet. Sci. Lett. 27, 108–112, 1975.

23 M.B. Steiner, Investigation of the geomagnetic field polarity during the Jurassic, J. Geophys. Res. 85, 3572–3586, 1980.

24 A.E.M. Nairn and T.J. Schmitt, A paleomagnetic study of the Ammonitico Rosso of Tunisia, in: Rosso Ammonitico Symposium Proceedings, A. Farinacci and S. Elmi, eds., pp. 471–478, Edizioni Tecnoscienza, Rome, 1981.

25 H.C. Soffel, Paleomagnetism of Upper Jurassic diabases and radiolarites from East Elba, EOS 60, 109, 1979.

26 J.G. Ogg, J.E.T. Channell, E.L. Winterer and P.O. Baumgartner, Magnetostratigraphy of Oxfordian and Kimmeridgian cherts and siliceous limestones of northern Italy, Int. Assoc. Geomagn. Aeron. Bull. 45, 209–210, 1981.

27 F. Heller, Rock magnetic studies of Upper Jurassic limestones from southern Germany, J. Geophys. 44, 525–543, 1978.

28 R.J.E. Johnson, R. Van der Voo and W. Lowrie, Paleomagnetism and late diagenesis of Jurassic carbonates from the Jura Mountains, Switzerland and France, Geol. Soc. Am. Bull. 95, 478–488, 1984.

29 J.E.T. Channell, Gross features of Jurassic geomagnetic stratigraphy from Umbrian and sections, EOS 61, 216, 1980.

30 J.E.T. Channell, J.G. Ogg and W. Lowrie, Geomagnetic polarity in the early Cretaceous and Jurassic, Philos. Trans. R. Soc. London, Ser. A 306, 137–146, 1982.

31 H. Tintant and P. Viallard, Le Jurassique moyen et superieur de la Chaine iberique sud occidentale aux confins des provinces de Teruel, Valencia et Cuenca, C.R. Somm. Soc. Geol. Fr. 6, 207–209, 1970.

32 A. Goy, G. Melendez, L. Sequeiros and J. Villena, El Jurasico superior del sector comprendido entre Molina de Aragon y Monreal del Campo (Cordillera Iberica), Cuad. Geol. Univ. Granada 10, 95–106, 1981.

33 L. Sequeiros and G. Melendez, Nuevos datos bioestratigraficos del Calloviense y Oxfordiense de Aguilon (Cordillera Iberica, Zaragoza), Guad. Geol. Univ. Granada 10, 167–177, 1981.

34 G. Melendez, E. Oxfordiense en el sector central de la Cordillera Iberica [S. Zaragoza–S. Tervel], Tesis doctoral, Dep. Paleontologia, Univ. Complutense (Madrid) and Univ. Zaragoza, 1984.

35 E. Cariou, R. Enay and H. Tintant, Oxfordien (Province submediterranéenne), in: Les zones du Jurassique en France. R. Mouterde et al., C. R. Somm. Seances Soc. Geol. Fr. 1971, 93–96, 1971.

36 W. Brochwicz-Lewinski, Early Oxfordian perisphinctids of the Czestochowa area; their stratigraphic value, Bull. Acad. Polonaise Sci., Ser. Sci. Terre 28, 233–242, 1980.

37 J.L. Simon Gomez, Reactivacion alpina del desgarre del Segre en el borde NE de la Cadena Iberica, Rev. Teruel 65, 195–209, 1981.

38 J.L. Kirschvink, The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc. 62, 699–718, 1980.

39 D.V. Kent and F.M. Gradstein, A. Jurassic to recent chronology, in: The geology of North America, in: The Western Atlantic Region, B.E. Tucholke and P.R. Vogt, eds., Geol. Soc. Am., Decade of North American Geology Ser. 1, in press, 1985.

40 D.V. Kent, F.M. Gradstein and F.P. Agterberg, A. Cretaceous and Jurassic geochronology, Geol. Soc. Am. Bull., in press, 1985.

41 W.B. Harland, A.V. Cox, P.G. Llewellyn, C.A.G. Pideton, A.G. Smith and R. Walters, A. Geologic Time Scale, 132 pp., Cambridge University Press, Cambridge, 1982.

42 G.E.G. Westermann, New Jurassic absolute dates and an improved method for stage duration estimates, in: Circum-Pacific Jurassic Research Group, IGCP #171, Rep. #2, G.E.G. Westerman, ed., pp. 86–88, McMaster University, Ontario, 1984.

43 G.E.G. Westermann, Gauging the duration of stages: a new approach for the Jurassic, Episodes 7/2, 26–28, 1984.

44 R.E. Sheridan, F.M. Gradstein et al., Init. Rept. DSDP 76, 947 pp., 1983.

45 S.C. Cande, D.V. Kent and J.L. LaBrecque, 1979, Interpreting paleomagnetic field behavior from marine magnetic anomalies, EOS 60, 236.

46 J. Vandenberg, New paleomagnetic data from the Iberian Peninsula, Geol. Mijnb. 59, 49–60, 1980.

47 J.-J. Schott, R. Montigny and R. Thuizat, Paleomagnetism and potassiumargon age of the Messejana Dike (Portugal and Spain): angular limitation to the rotation of the Iberian Peninsula since the Middle Jurassic, Earth Planet. Sci. Lett. 53, 457–470, 1981.

48 P. Viallard, La Chaine Iberique: zone de cisaillement intracontinentale pendant la tectogenèse alpine, C. R. Acad. Sci. Paris 289, 65–68, 1979.